Anjali Amgain

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Professional Summary

Seeking a full-time position in Traffic or Transportation Engineering where I can apply hands-on experience with traffic modeling tools to deliver data-driven intersection and signal timing solutions.

Education

Southern Illinois University Edwardsville — Edwardsville, IL
Master of Science in Civil Engineering. Specializations: Transportation Engineering, Traffic Engineering | FE Exam - Expected October 2026. — *Anticipated December 2026*

Relevant Coursework: Transportation Safety Systems, Traffic Signals, Transportation Planning, Traffic Studies, Municipal Infrastructure, Sustainable Engineering, Technical and Business Writing

Experience

Graduate Research Assistant — February 2026 – Present
Southern Illinois University Edwardsville — *Edwardsville, IL*
- Analyzed spatial failure data for traffic cameras and dynamic message signs across the IDOT District 8 network using ArcGIS and Asset Inventory.
- Co-authored a peer-reviewed paper on GIS and Weibull survival analysis of IDOT District 8 traffic cameras, accepted for the 2027 TRB Annual Meeting.

Seasonal Engineering Intern — May 2026 – August 2026
Illinois Department of Transportation — *Colona, IL*
- Inspected construction, documented quantities, and provided field-office support for roadway projects.
- Evaluated highway work zones and traffic control plans for compliance with MUTCD.

Graduate Teaching Assistant — August 2025 – December 2025
Southern Illinois University Edwardsville — *Edwardsville, IL*
- Led undergraduate lab activities focused on traffic modeling and analytical problem-solving.
- Evaluated transportation planning coursework covering transportation systems and engineering design.

Sub Engineer — June 2024 – August 2025
Likhu Rural Municipality, Government of Nepal — *Nuwakot, Nepal*
- Surveyed sites and implemented designs while maintaining construction quality control.
- Prepared project completion reports to secure government approval.

Projects

Academic Projects
- Modeled the Troy Rd corridor in Edwardsville using VISSIM for the Troy Rd Traffic Signal Coordination project, improving LOS from D to B through signal timing and transit-priority optimization across three intersections: Montclaire Ave, Harvard Dr., and Edwardsville Crossing Dr.
- Analyzed peak-hour and post-COVID congestion in Madison and Cook Counties using GIS density tools.
- Assessed ITS reliability by fitting Weibull curves to 2021-2026 camera and DMS uptime data.

Skills

Software: Civil 3D, SYNCHRO, VISSIM, VISUM, ArcGIS Pro, AutoCAD, OpenRoad, MicroStation
Documentation: Technical report preparation, Construction documentation

Certifications

- BDE Geometric Design Training (IDOT)
- Civil 3D: Basic Roadway Design

Awards & Honors

- Graduate Scholar Award 2025
- ECF Disaster Preparedness Scholarship

Activities & Leadership

ITE SIUE Student Chapter: *Vice President* — August 2026- Present
American Society of Civil Engineers (ASCE): *Student Member* — August 2025-Present
Society of Women Engineers: *Student Member* — August 2025- Present
SIUE'S Dining Services: *Starbucks Barista* — February 2026- Present